Sickness and Health

The desire to stay alive and healthy is a basic human instinct. It is no wonder, then, that people in early civilizations attempted to explain the origins of disease—and intervened to soothe pain, encourage healing, and effect cures. Some of these traditional approaches to medicine are still in use today.

« BEFORE

Archaeological evidence suggests that prehistoric hunter-gatherers were robust and healthy. Their diets—mainly raw fruits, leaves, and vegetables, with some lean meat and fresh fish—were probably very well attuned to the needs of the human body.

MOVING AROUND

As they were constantly on the move, and not living in large groups, hunter-gatherers probably **suffered rarely from infectious diseases**. Life expectancy was low, but this probably had more to do with physical dangers than disease or want.

FLINT DRILL

MEDICAL INTERVENTION

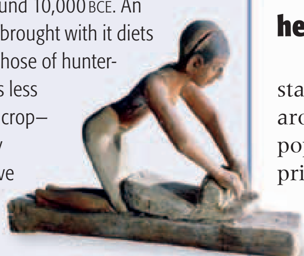
During the Neolithic period, people began to make sophisticated stone tools and weapons. Some tools were used in **primitive attempts at surgery**. For example, flint-tipped dental drills, found in Pakistan, date back as far as 7000 BCE. Teeth in remains found nearby showed signs of skilful drilling to remove rotten dental tissue.

LIVING TOGETHER

As people began domesticating animals and crops, communities grew larger, and their inhabitants started living more closely together **« 38–39**. They lived close to their livestock, too, and this led to a **proliferation of diseases** that had not been a problem before. Waste was another hazard, and water supplies quickly became contaminated.

CHANGING DIETS

Settled, or sedentary, farming first appeared in the **Fertile Crescent** **« 36** in what is now the Middle East, around 10,000 BCE. An agricultural lifestyle brought with it diets very different from those of hunter-gatherers. There was less variety, and a single crop—often wheat—usually dominated. Repetitive tasks, such as grinding grain to make flour, caused excessive wear to people's joints, leading to arthritis. At the same time, **more food was cooked**, a process that can destroy vitamins and introduce toxins, while babies depended less on their mothers' milk. These changes led to populations of smaller stature and with weaker bones, as well as new conditions such as **anemia and scurvy**.



GRINDING GRAIN

EARLY HEALERS

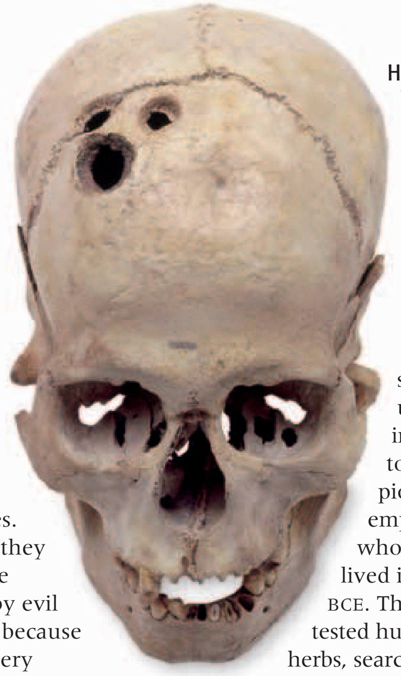
Other medical interventions were practiced besides dentistry (see above). Often, serious bone fractures were successfully reset—remains show signs of regrowth. And in caves at Lascaux, France, archaeologists have found preparations of **medicinal herbs** dating back to 13,000 BCE.

At the beginning of the Bronze Age, around 3000 BCE, the civilizations of Mesopotamia (see pp.54–55), ancient Egypt (see pp.64–73), and the Indus Valley (see pp.58–59) were already well established. Busy, booming cities were surrounded by fertile land given over to agriculture, and farming became so efficient that only a relatively small proportion of the population needed to be involved in producing food. This led to the development of trade, mathematics and astronomy, writing, and a flourishing of cultural activities.

The price of progress

Along with the many benefits of their way of life, however, people in early civilizations suffered from some ill effects. Their diets were generally lower in fiber and higher in fat and salt than their hunter-gatherer predecessors. There is evidence that this led to an increase in conditions such as high blood pressure, heart disease, and cancer—a trend that began with the rise of agriculture several thousand years earlier, and continues today. This pattern was repeated elsewhere. In Central America, for example, early Maya people began relying on corn as a staple in civilizations originating around 1000 BCE. This led to a population explosion, but at the price of a dangerously restricted diet.

As the Bronze Age gave way to the Iron Age in Europe and Asia after about 1000 BCE, many killer diseases arose for the first time in human populations. Smallpox and anthrax are two good examples. In both cases, and in many others, the pathogens (disease-causing organisms) evolved to cross species barriers from livestock, and were able to take hold because people were living so close together in mostly unsanitary conditions. Rats, fleas, and lice thrived, and carried diseases such as the plague and typhoid. In times of flood, drought, or war, these problems were heightened.



Hole in the head
The earliest known surgery, dating back to 40,000 BCE, was trepanning—drilling a hole in the skull. This was probably done to release evil, disease-causing demons. The practice occurred in Central America, Europe, and Asia.

Explaining disease

No one in early civilizations could understand disease the way modern medical science does. Thus, it was normal to attribute the causes of disease to supernatural forces. People believed that they became unwell as the result of possession by evil spirits—demons—or because of angry gods or sorcery carried out by their enemies.

Just as explanations of disease appealed to the supernatural, so did most attempts to cure people. In most cultures, priests and sorcerers were at least as important as physicians—and exorcism of demons, sacrifice to the gods, shamanistic rituals, and counter-sorcery were commonplace. The Ebers papyrus, written in Egypt in the 2nd millennium BCE and discovered in the 19th century, contains a long list of "medical" incantations designed to turn away evil spirits.

Egyptian surgical instruments
In ancient Egypt, sharp bronze and copper instruments were used when embalming the dead, as well as for operating on the living.

Herbalism

Healing based on supernatural beliefs is an example of folk medicine. Herbal remedies also fall into this category. Many ancient treatments based on herbs or other plants evolved through trial and error, and are so successful that they are still used today for their analgesic (pain relief), antibiotic, or antifungal action. In Mesopotamia, for example (see pp.54–55), a willow bark extract was used to relieve headaches and reduce fevers. That extract is salicylic acid, the basis of aspirin.

China has the strongest tradition of using herbs and roots in medicine. According to legend, one of its pioneers was the emperor Shen Nong, who is supposed to have lived in the 3rd millennium BCE. The story goes that he tested hundreds of different herbs, searching for ones with medicinal effects. He is also credited with the introduction of tea-drinking—for its remedial qualities.

The therapeutic use of plants is found in almost every corner of the world. The Olmec in Central America, for example (see pp.74–75), had areas of their gardens set aside for growing medicinal herbs. Papyri from ancient Egypt list remedies involving plants such as thyme, juniper, frankincense, and garlic—although there were others that used beer and animals' entrails. Herbalism is also central to Ayurvedic medicine, which originated in the Vedic period of India (see p.144) shortly before 1000 BCE. Ayurveda (literally "knowledge of life") is a holistic system that uses a combination of religion and science to create physical, mental, and spiritual well-being.

Organized approach

The Ayurvedic system is typical of the approach to science and technology that began to emerge—in China and India in particular—during the 1st millennium BCE. People began to think rationally, organize their thoughts, discuss them with others, and derive theories. This approach led not only to an encyclopedic knowledge of human anatomy and of a vast range of diseases, but also to well thought-out systems of diagnosis and treatment—the basis of modern medicine.